

Backend Verification Report

CodeSpec AI | Run date: 2026-10-03

Overall test result

PASS: 50 tests passed in the host suite and in the rebuilt Docker image. Each has two non-failing warnings: Starlette/httpx deprecation and local-Qdrant payload-index notice.

Focused tests

Area	Result
Ingestion	6 passed in 2.11s
Tree-sitter parsing	21 passed in 0.68s
Embedding, Qdrant, and chunking	14 passed, 1 warning in 1.19s
Relationship extraction and Neo4j impact	3 passed in 0.68s
FastAPI and Celery	7 passed, 1 warning in 1.41s
GraphRAG / hybrid retrieval	3 passed in 0.58s
Live FastAPI	PASS: /health reports ok; vector_store ready
Docker data services	PASS: Redis, Neo4j, and Qdrant healthy
Celery queues	PASS: celery and embeddings workers respond pong
Docker Compose	PASS: config parses; Qdrant healthcheck fixed and verified

Per-area commands and evidence

1. Ingestion

Main files: app/workers/tasks.py; app/core/ingestion/scanner.py; git_fetcher.py; zip_handler.py

Command: python -m pytest tests/test_module2_tasks.py tests/test_index_workspace.py tests/test_repository_indexing.py -q --tb=short

Output: 6 passed in 2.11s (full transcript: ingestion.txt)

2. Tree-sitter parsing

Main files: app/core/parser/registry.py; app/core/parser/; app/models/parser_models.py

Command: python -m pytest tests/test_module1.py -q --tb=short

Output: 21 passed in 0.68s (full transcript: tree-sitter.txt)

3. Embedding, Qdrant, and chunking

Main files: app/core/embeddings/service.py; app/core/vectors/indexer.py; app/core/vectors/store.py; app/core/chunking/service.py

Command: python -m pytest tests/test_embeddings.py tests/test_vector_store.py tests/test_chunking.py -q --tb=short

Output: 14 passed, 1 warning in 1.19s (full transcript: embedding-qdrant-chunking.txt)

4. Relationship extraction and Neo4j impact

Main files: app/core/graph/builder.py; app/db/neo4j.py; app/core/impact/service.py

Command: python -m pytest tests/test_impact_service.py -q --tb=short

Output: 3 passed in 0.68s (full transcript: relationship-neo4j-impact.txt)

5. FastAPI and Celery

Main files: app/main.py; app/api/routes/; app/workers/celery_app.py; app/workers/tasks.py

Command: python -m pytest tests/test_module2_api.py tests/test_module2_tasks.py -q --tb=short

Output: 7 passed, 1 warning in 1.41s (full transcript: fastapi-celery.txt)

6. GraphRAG / hybrid retrieval

Main files: app/core/retrieval/service.py; app/core/retrieval/graph.py; app/api/routes/retrieval.py; app/api/routes/rag.py

Command: python -m pytest tests/test_hybrid_retrieval.py -q --tb=short

Output: 3 passed in 0.58s (full transcript: graphrag.txt)

Live service outputs

FastAPI GET /health:

```
{
  "status": "ok",
  "app": "CodeSpec AI",
  "version": "0.1.0",
  "vector_store": "ready"
}
```

Live Qdrant and Neo4j integration:

```
Qdrant live filtered upsert/search: PASS (1 hit)
Qdrant smoke collection cleanup: PASS
GraphBuilder parsed source files: 1
Neo4j caller-CALLS-target edges: 1
Neo4j graph smoke cleanup: PASS
```

Redis returned PONG; Neo4j returned 1 for RETURN 1 and persisted one caller-CALLS-target edge; Qdrant /healthz passed and repository-filtered vector search returned one hit. See docker-infrastructure.txt and runtime-smoke.txt for captured output.

Celery inspect ping returned pong for both live workers. Active queues were celery and embeddings. The API Swagger endpoint returned HTTP 200. See celery-workers.txt.

Docker build and configuration

The backend image built with `INSTALL_EMBEDDING_DEPS=false`; all 50 tests passed inside that image. The Qdrant client is pinned to 1.13.3 for the Compose server 1.13.2. The Compose healthcheck no longer calls curl inside the minimal Qdrant image; its process check and the actual HTTP endpoint were both verified.

Limitations: BGE-M3 model inference was not run because the heavyweight embedding dependency group and model weights were not provisioned. The embeddings queue worker was started and pinged, but no indexing task was submitted. The project-local backend .env file is absent; infrastructure ran through Compose, while API and Celery containers used explicit development environment values.

Saved evidence

This folder contains focused pytest transcripts, full host and Docker test-suite outputs, live service checks, the runtime smoke-test script and output, rendered Compose config, build log, and test-summary.json.